

Ethanol Blending in Gasoline – Vietnam

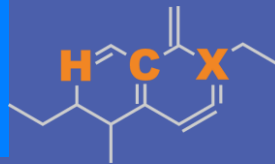
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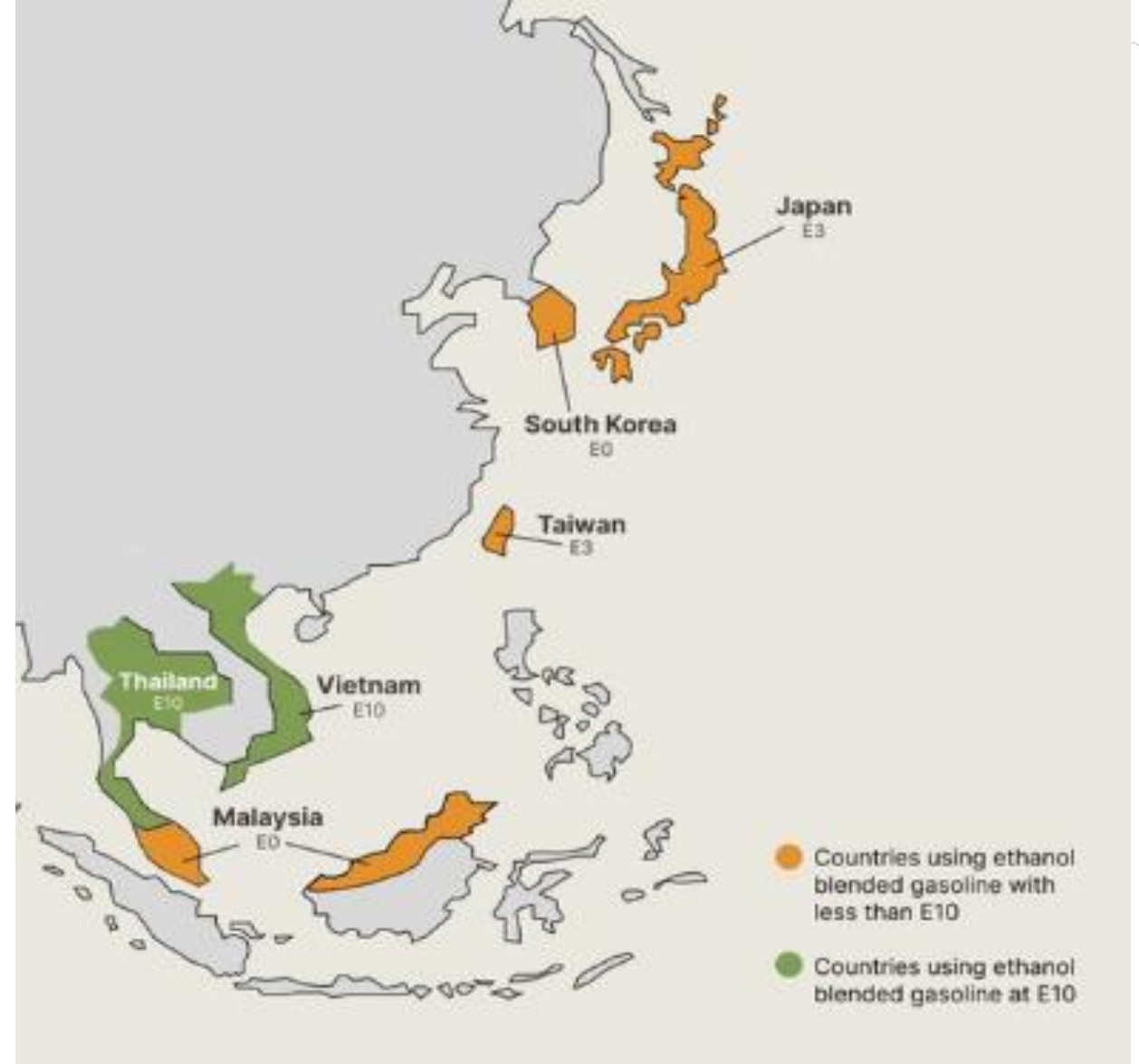


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Ethanol Blending in Asia

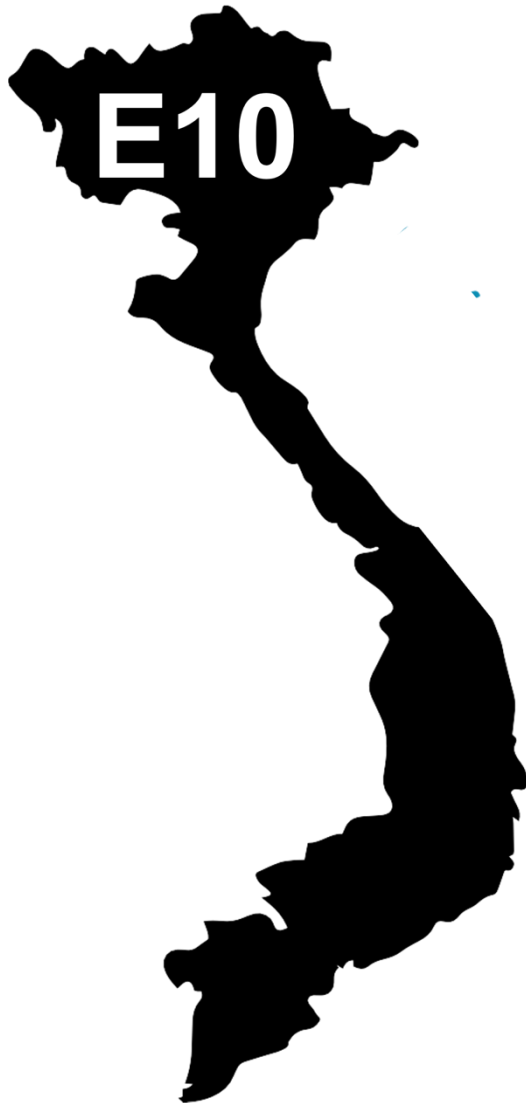
There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Vietnam



In 2024, gasoline consumption in Vietnam was 10,600 million liters (2,800 million gallons) and growth rate was 3.8%. Gasoline production and net imports were 7,870 and 2,730 million liters (2,079 and 721 million gallons) correspondingly. Vietnam complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 94.8 RON.

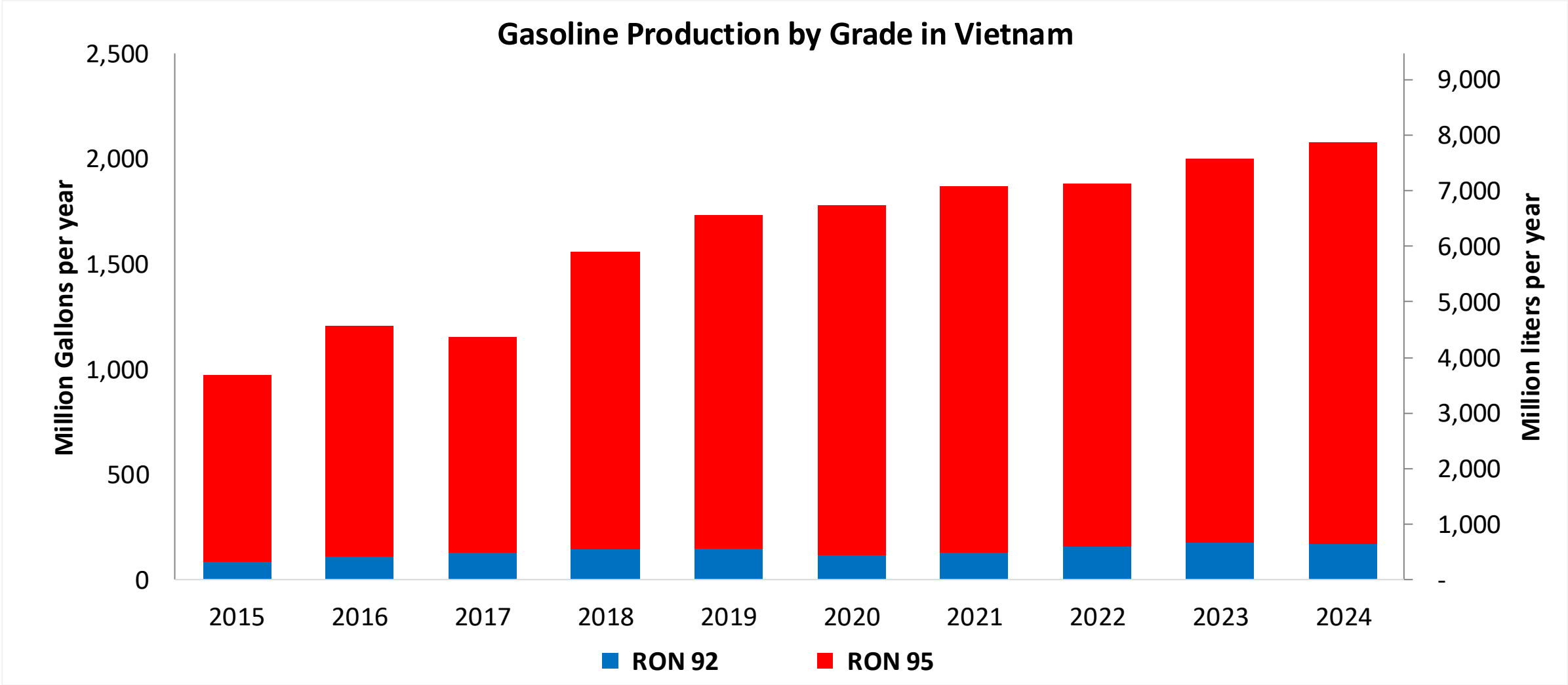
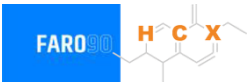
Vietnam's bioethanol demand is growing rapidly, driven by government policies like a planned nationwide E10 gasoline mandate set to begin in 2026. Ethanol consumption in Vietnam was 236 million liters (62 million gallons) representing 2.5% of gasoline demand. Ethanol production and Net Imports were 200 and 36 million liters (62 and 9 million gallons) correspondingly.

Source: Petro Vietnam, EIA,, EITdata, USDA

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). There are also methyl groups attached to the ring.

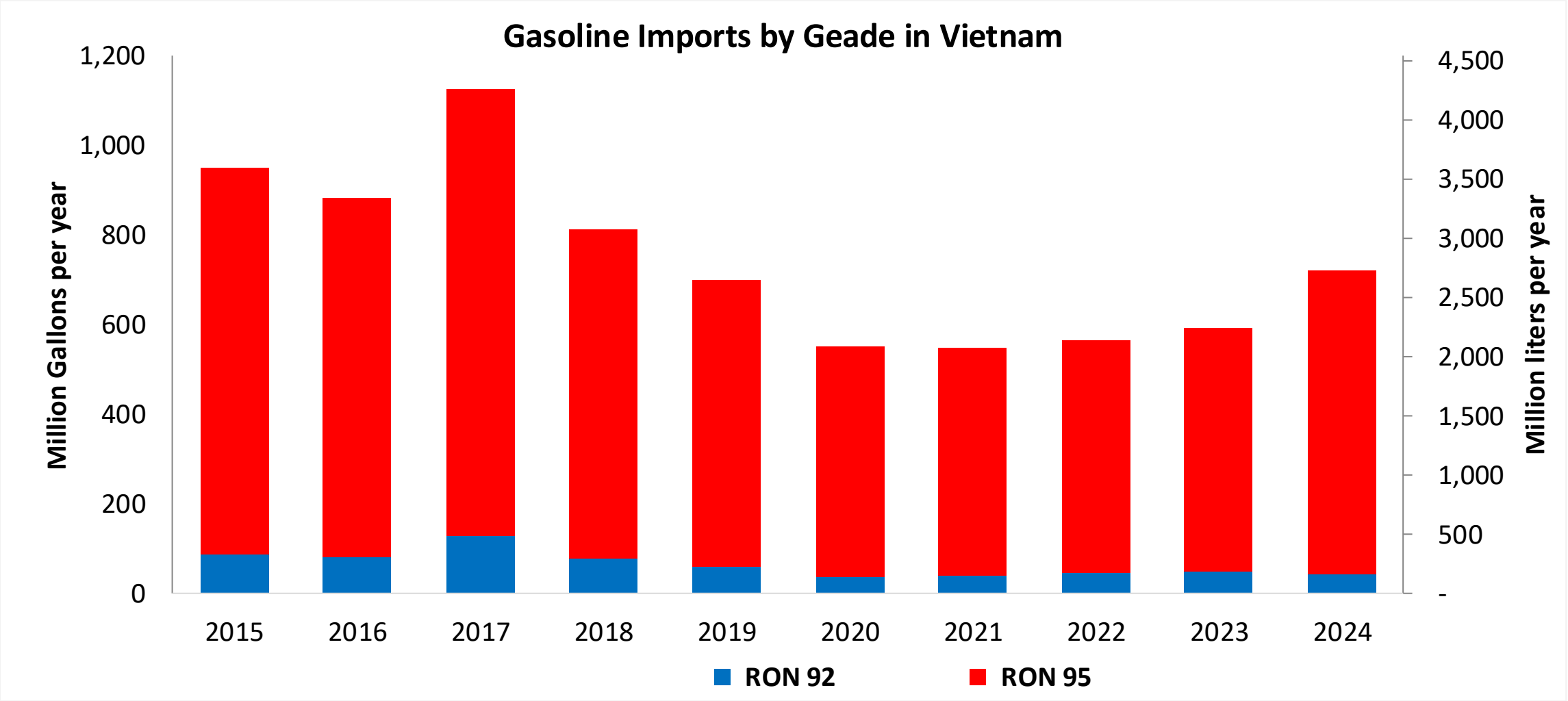


Gasoline Production in Vietnam



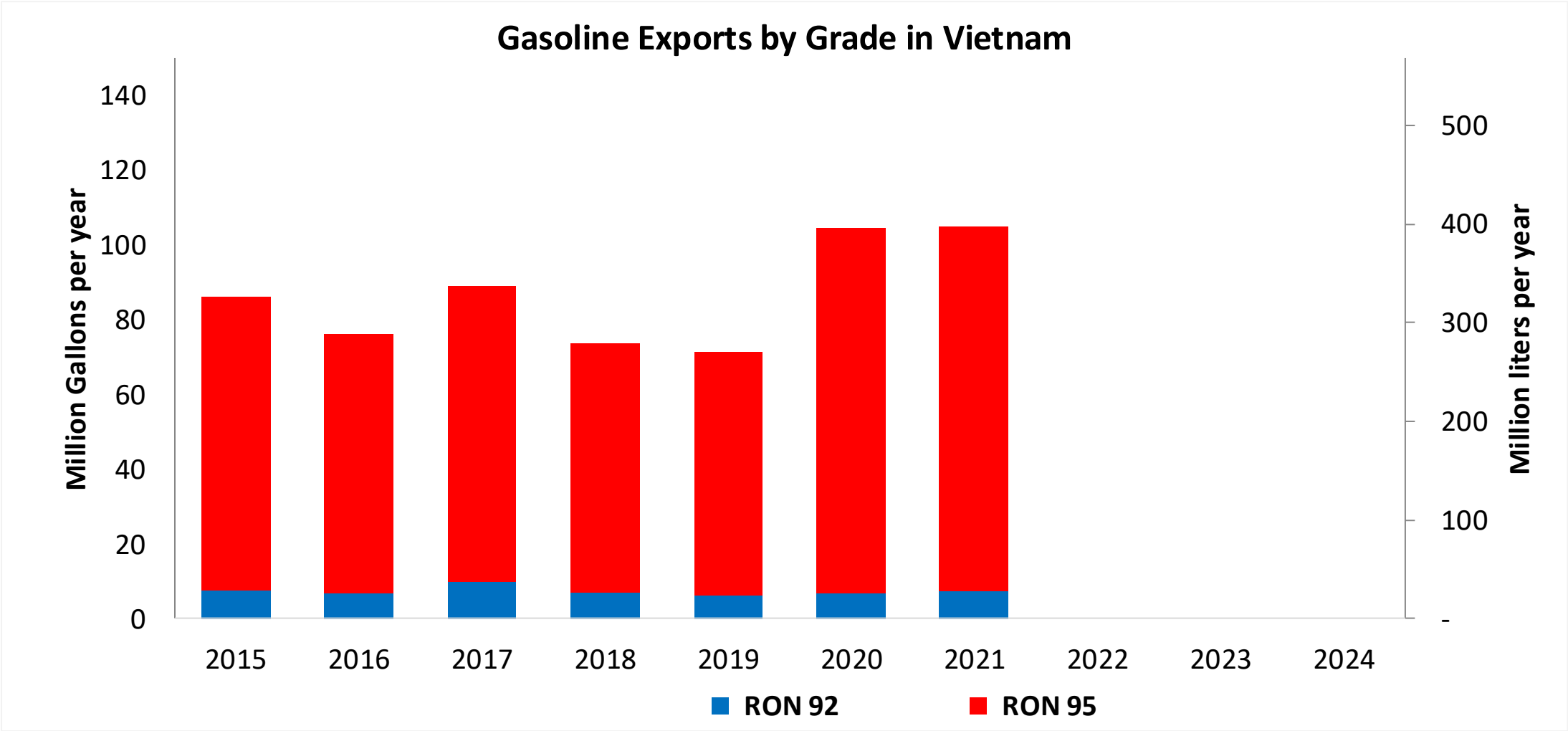
Source: Petro Vietnam, EIA,, EITdata

Gasoline Imports in Vietnam



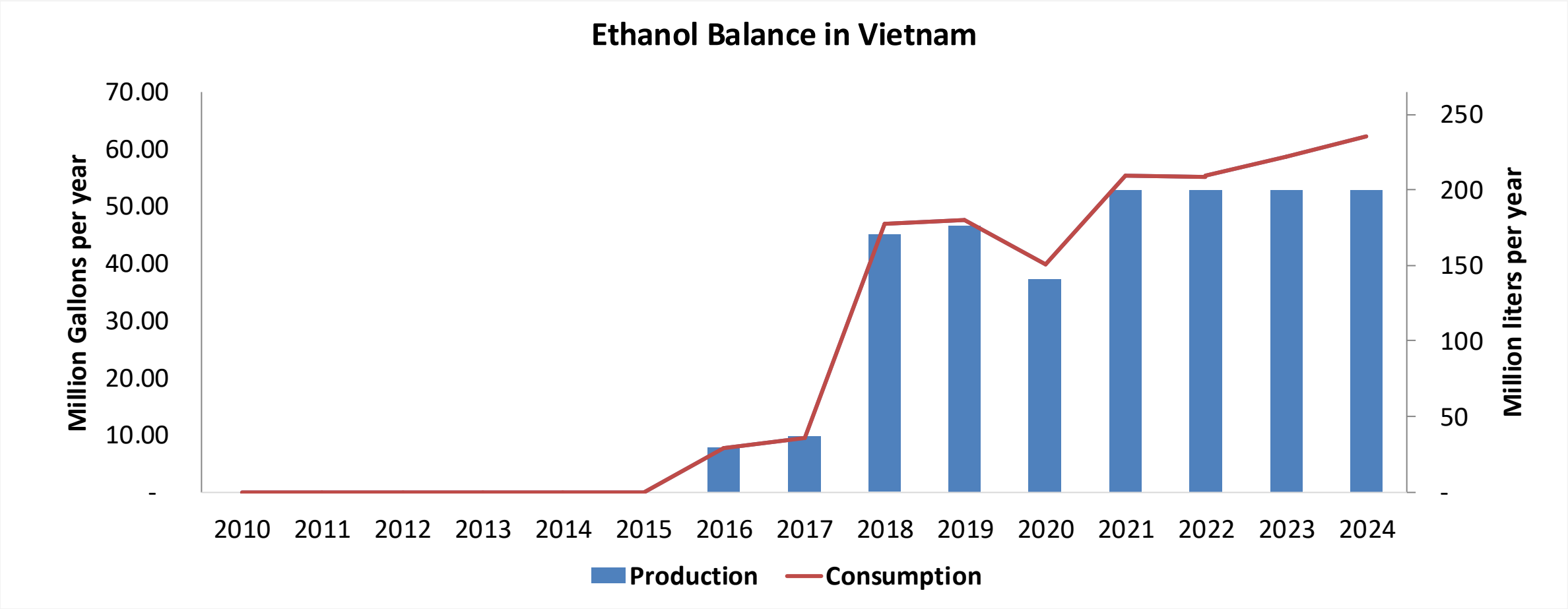
Source: Petro Vietnam, EIA, EITdata

Gasoline Exports in Vietnam



Source: Petro Vietnam, EIA,, EITdata

Ethanol Balance in Vietnam



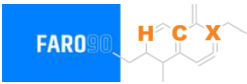
Source: Petro Vietnam, USDA

Gasoline Quality in Vietnam

Name	QCVN 01:2022/BKHCN					
Implementation Year	2023					
Applicability	All territory					
Grade	RON 92		RON 95		RON 97	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		97.0	
MON	-		-		-	
AKI	-		-		-	
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Manganese (mg/L)		5.0		5.0		5.0
Sulfur (ppm)		10		10		10
Oxygen (%w)		2.7		2.7		2.7
Ethanol (%vol)		-		-		-
MTBE (%vol)		-		-		-
RVP (psi)		8.7		8.7		8.7

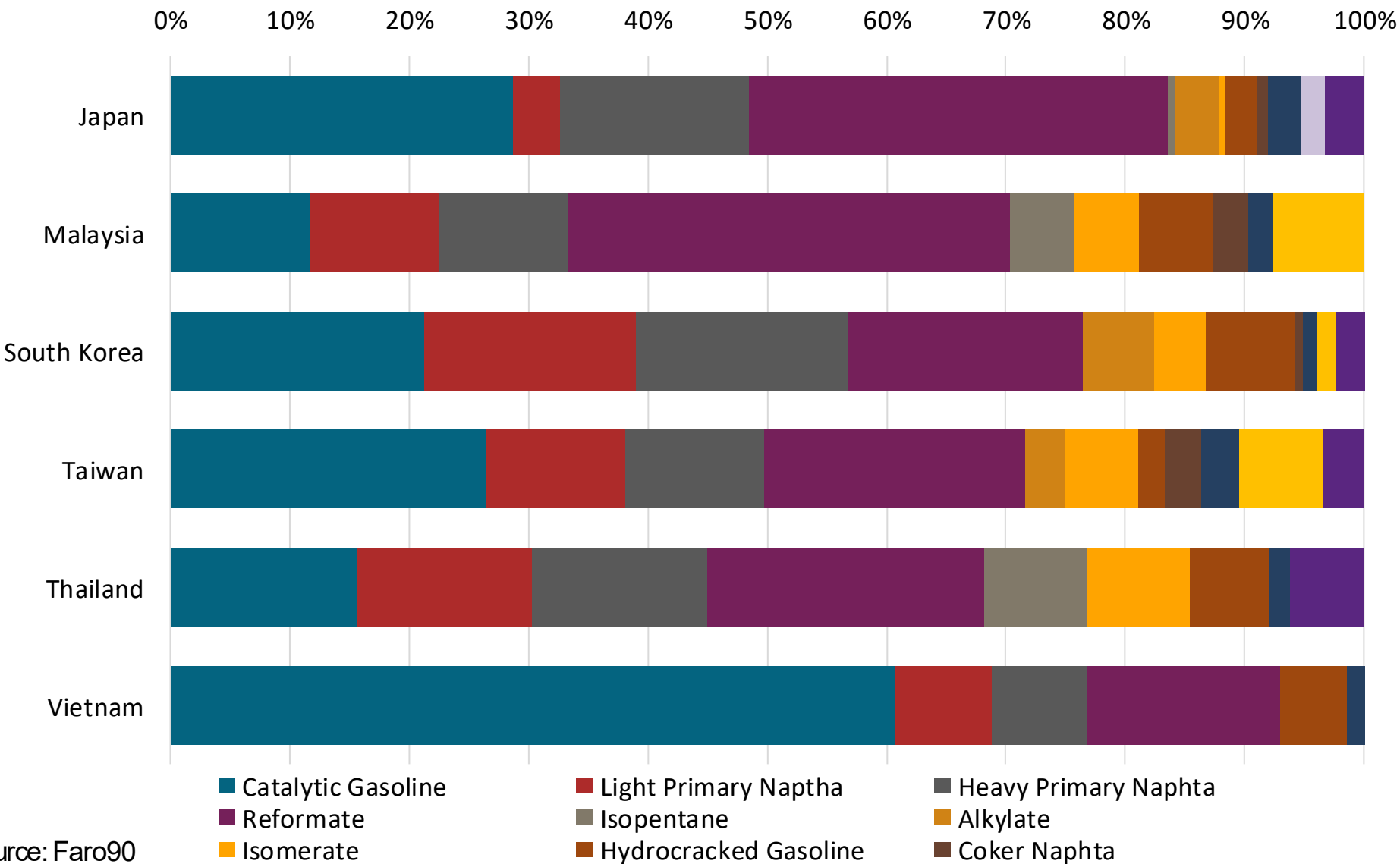
Source: Caselaw

Gasoline Component Blending in Asia



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

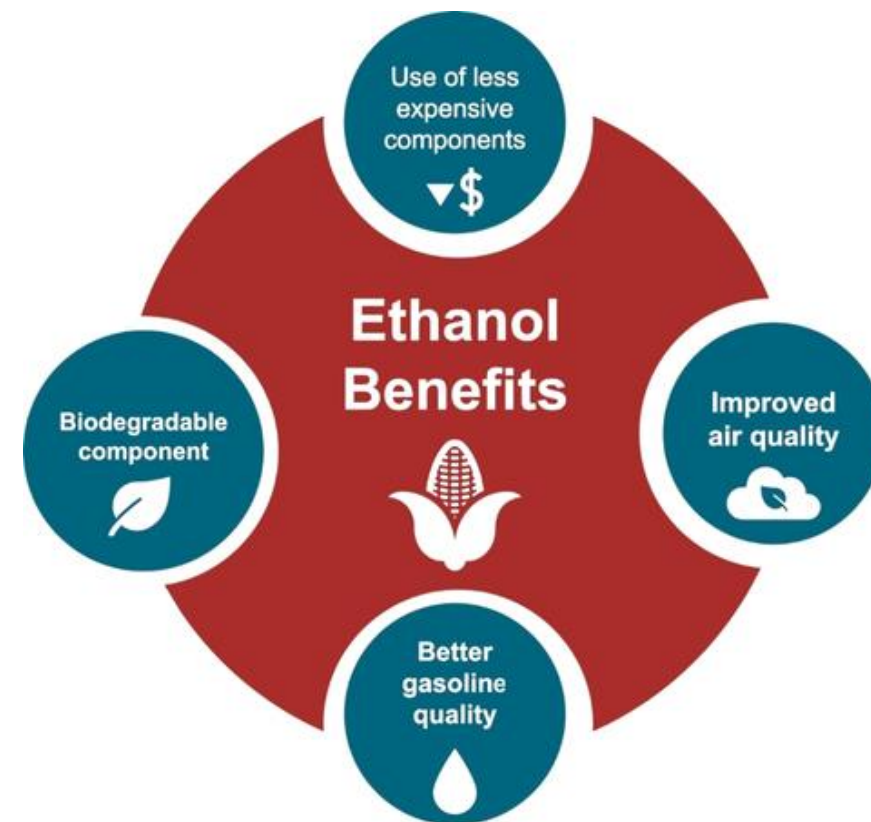
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

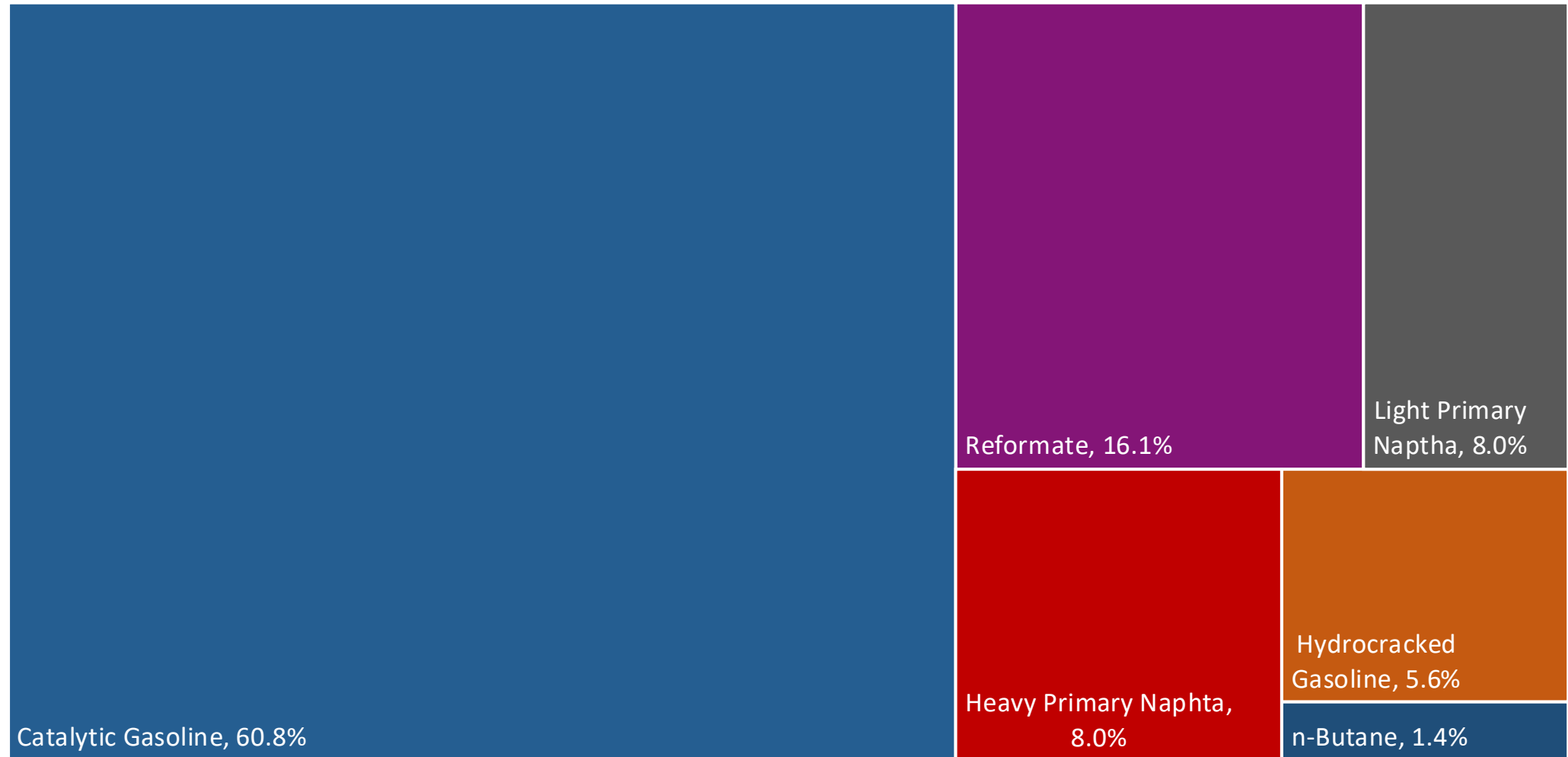
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

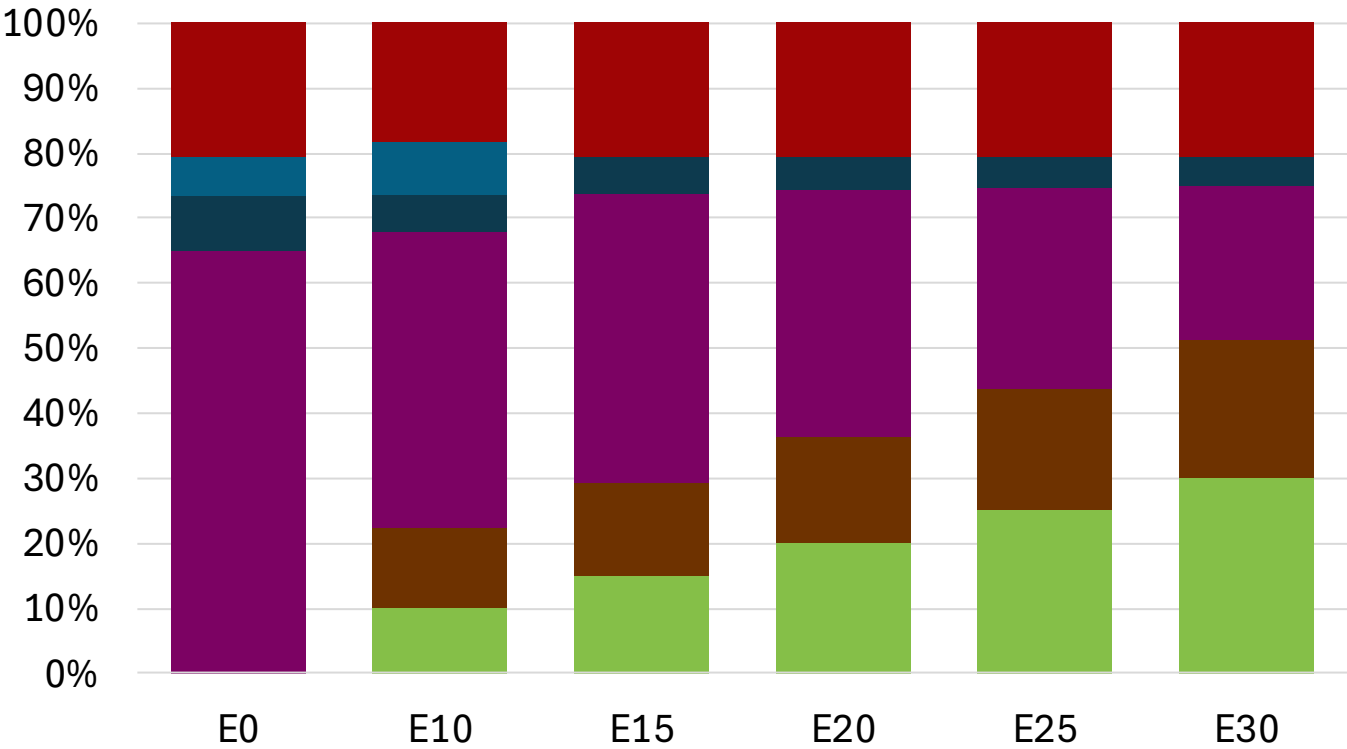


Vietnam Available Blending Components



Source: Faro90

Vietnam – Gasoline 92 – Constant Octane

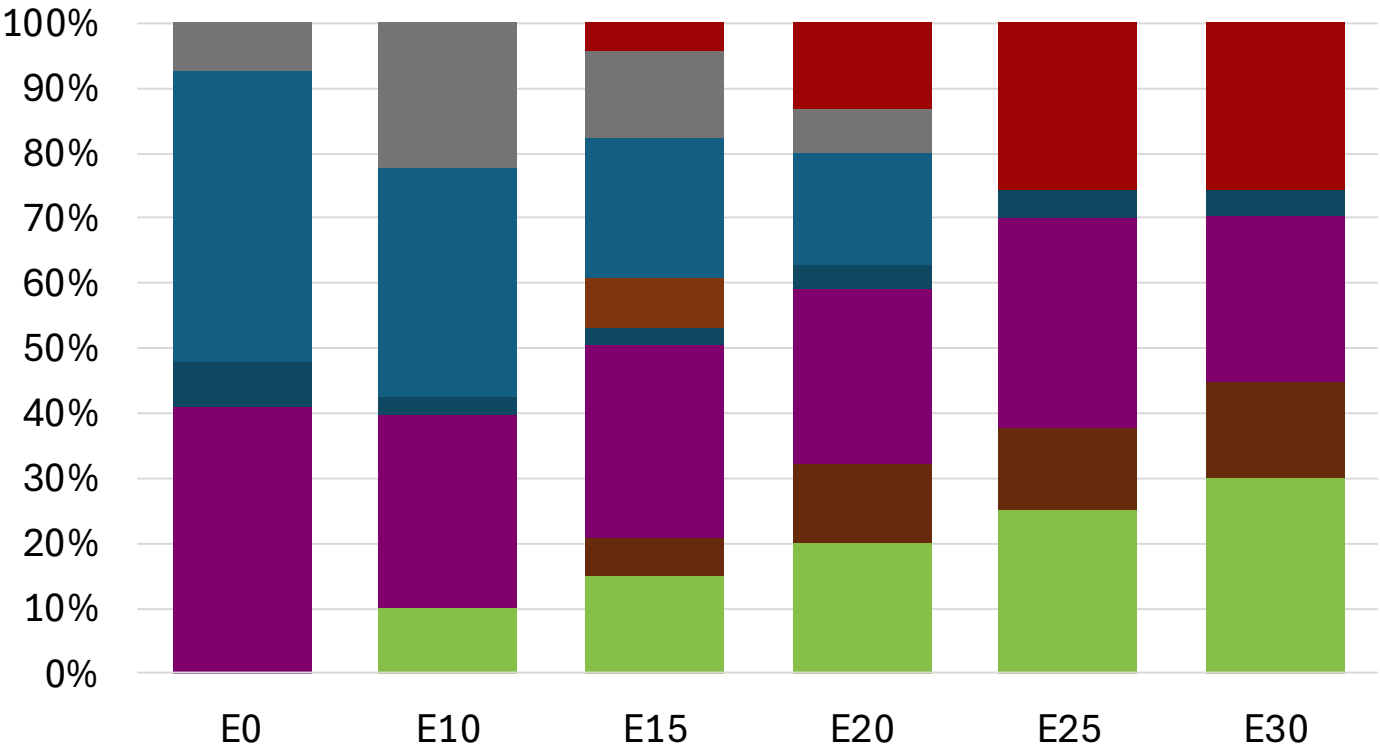


Average CIF 2024
Prices

Octane (RON)	92.0	92.0	92.9	94.0	95.0	96.0
Price (US\$/gal)	\$ 2.280	\$ 2.113	\$ 2.063	\$ 2.020	\$ 1.972	\$ 1.921

Source: Faro90

Vietnam – Gasoline 95 – Constant Octane

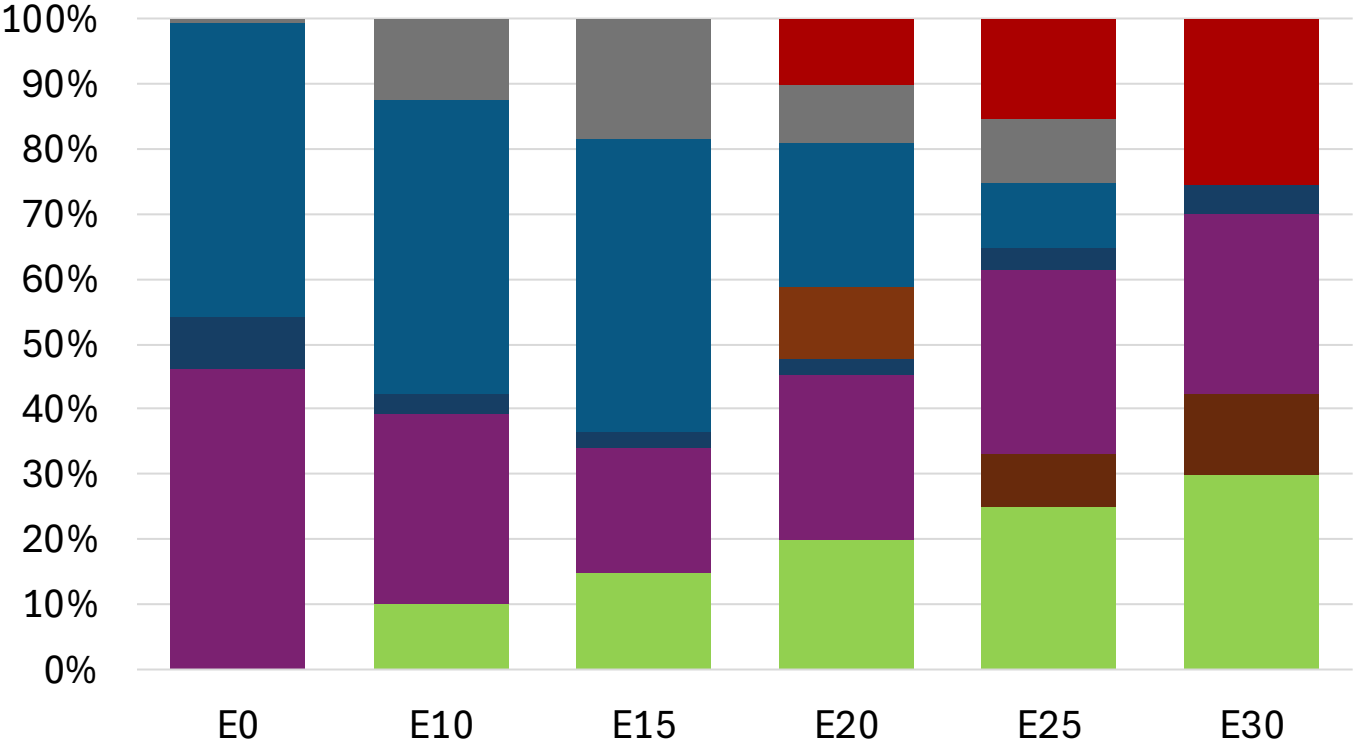
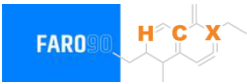


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.2	96.3
Price (US\$/gal)	\$ 2.436	\$ 2.305	\$ 2.214	\$ 2.090	\$ 1.981	\$ 1.933

Source: Faro90

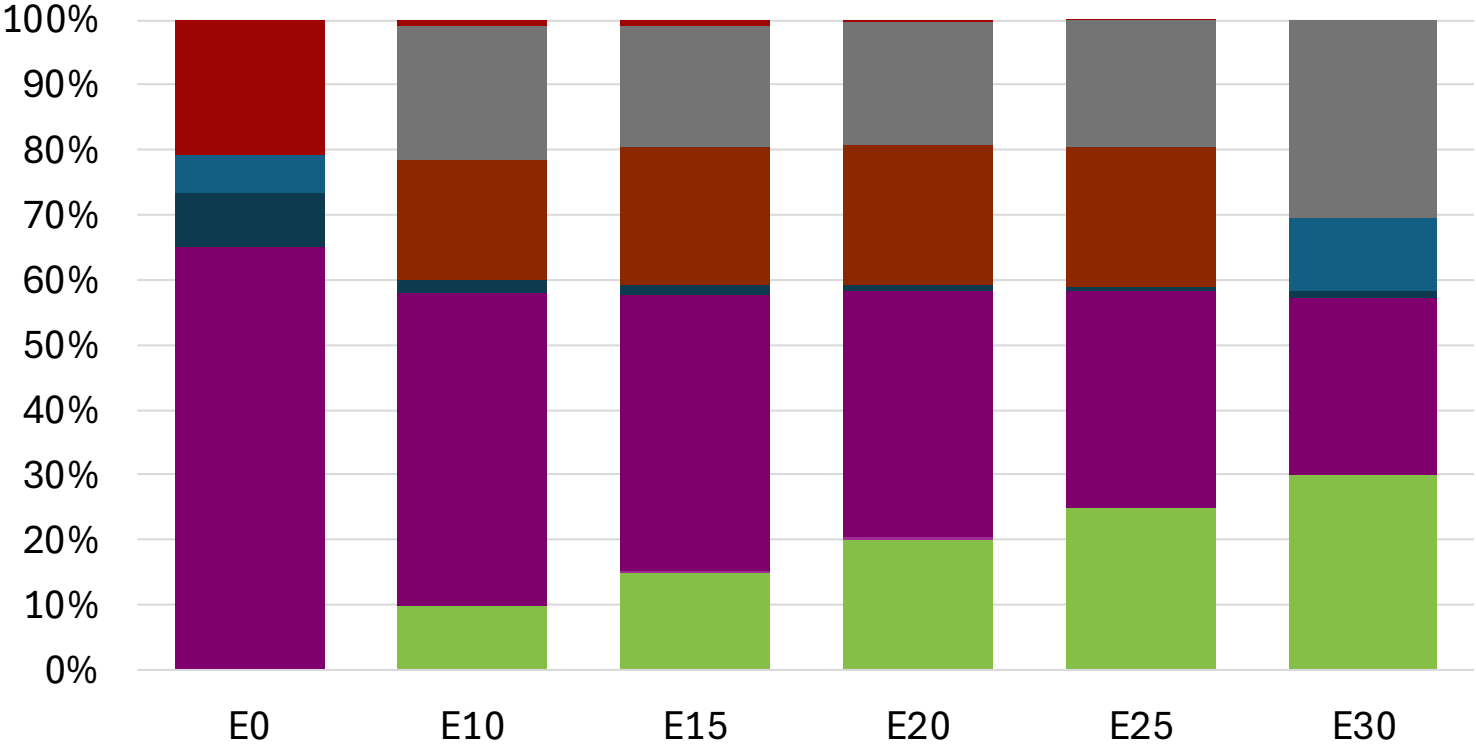
Vietnam - Gasoline 97 - Constant Octane



Average CIF 2024
Prices

Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.542	\$ 2.416	\$ 2.322	\$ 2.228	\$ 2.105	\$ 1.983

Vietnam – Gasoline 92 – Increased Octane

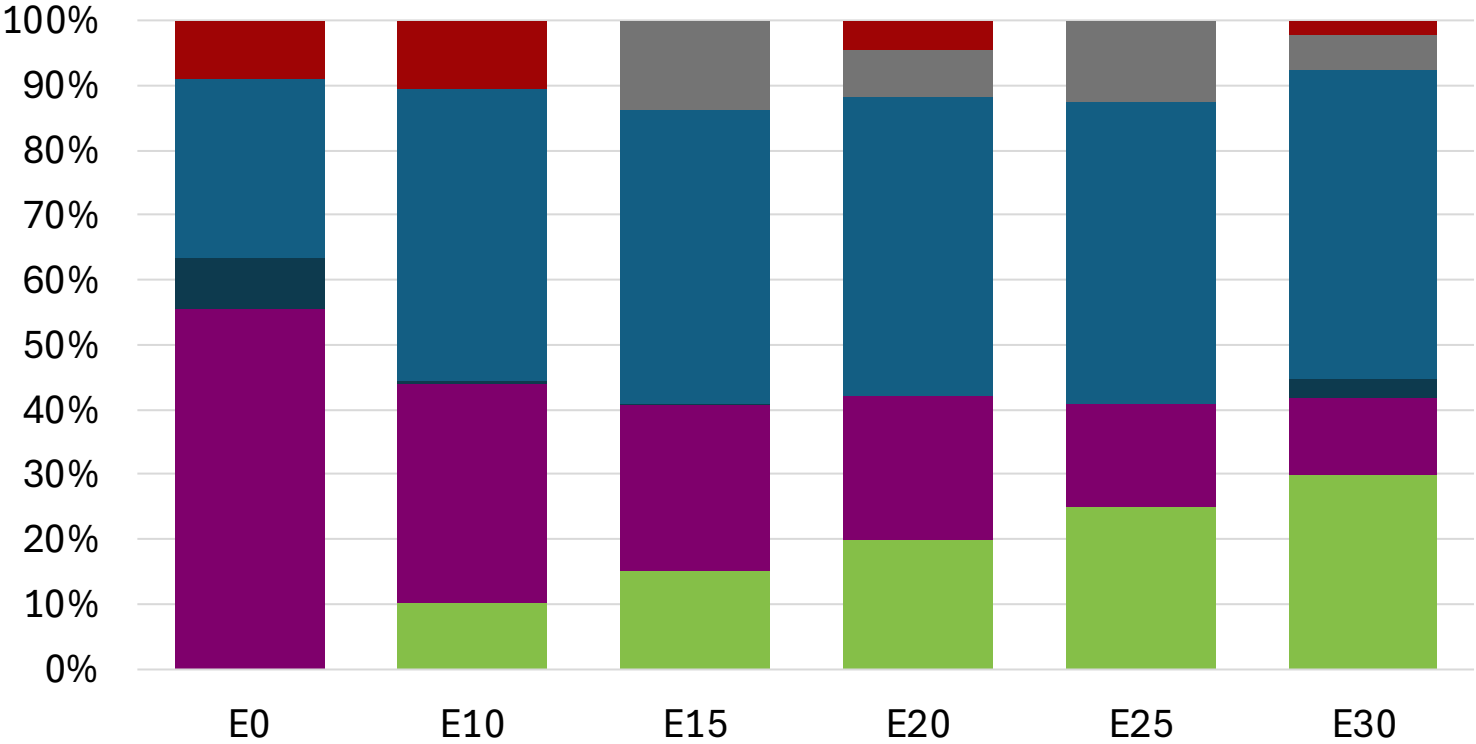


Average CIF 2024
Prices

Octane (RON)	92.0	94.1	95.7	97.4	99.0	101.0
Price (US\$/gal)	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280

Source: Faro90

Vietnam – Gasoline 95 – Increased Octane

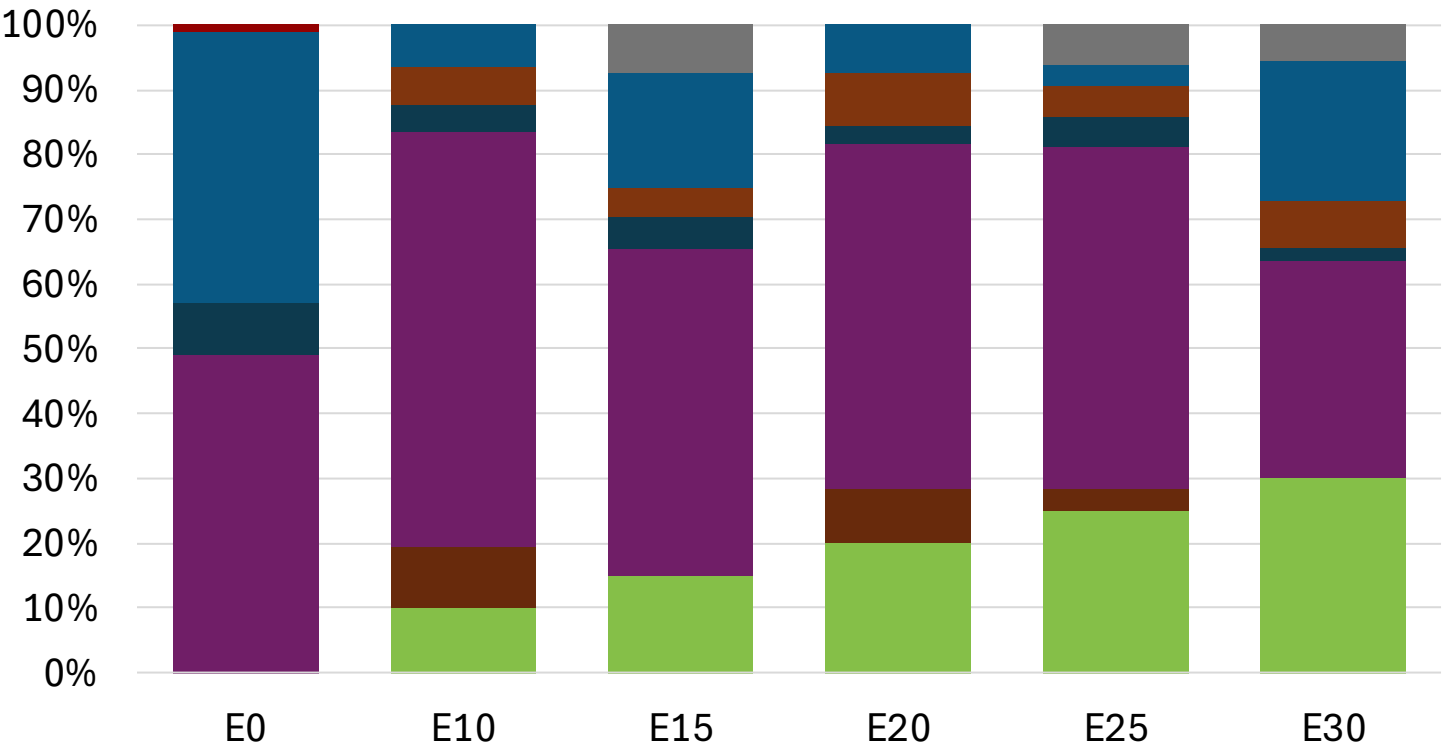


Average CIF 2024
Prices

Octane (RON)	95.0	96.8	98.4	100.0	101.6	103.7
Price (US\$/gal)	\$ 2.436	\$ 2.436	\$ 2.436	\$ 2.436	\$ 2.436	\$ 2.436

Source: Faro90

Vietnam – Gasoline 97 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	97.0	99.1	101.1	101.9	103.9	104.9
Price (US\$/gal)	\$ 2.541	\$ 2.541	\$ 2.541	\$ 2.541	\$ 2.541	\$ 2.541

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

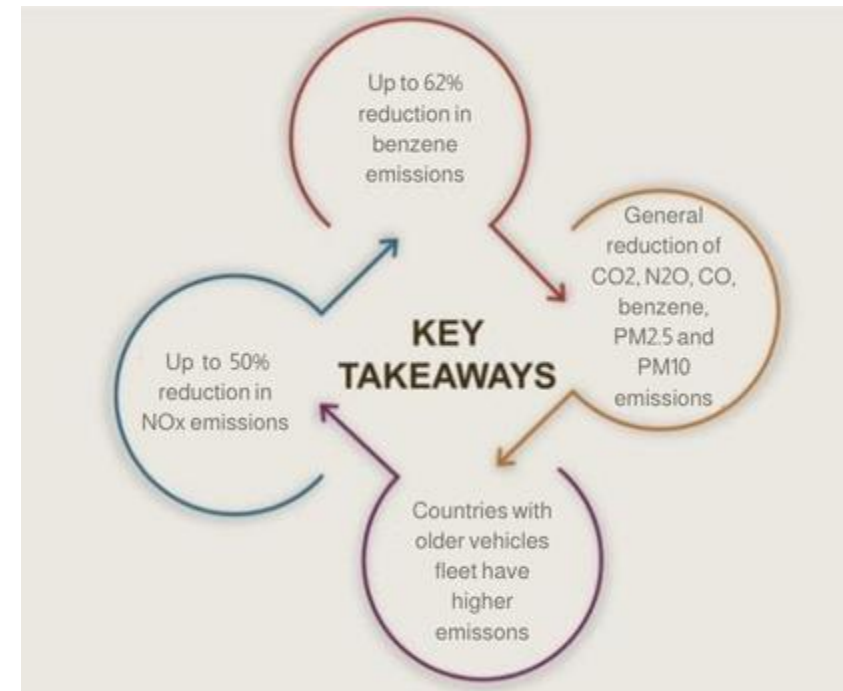
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

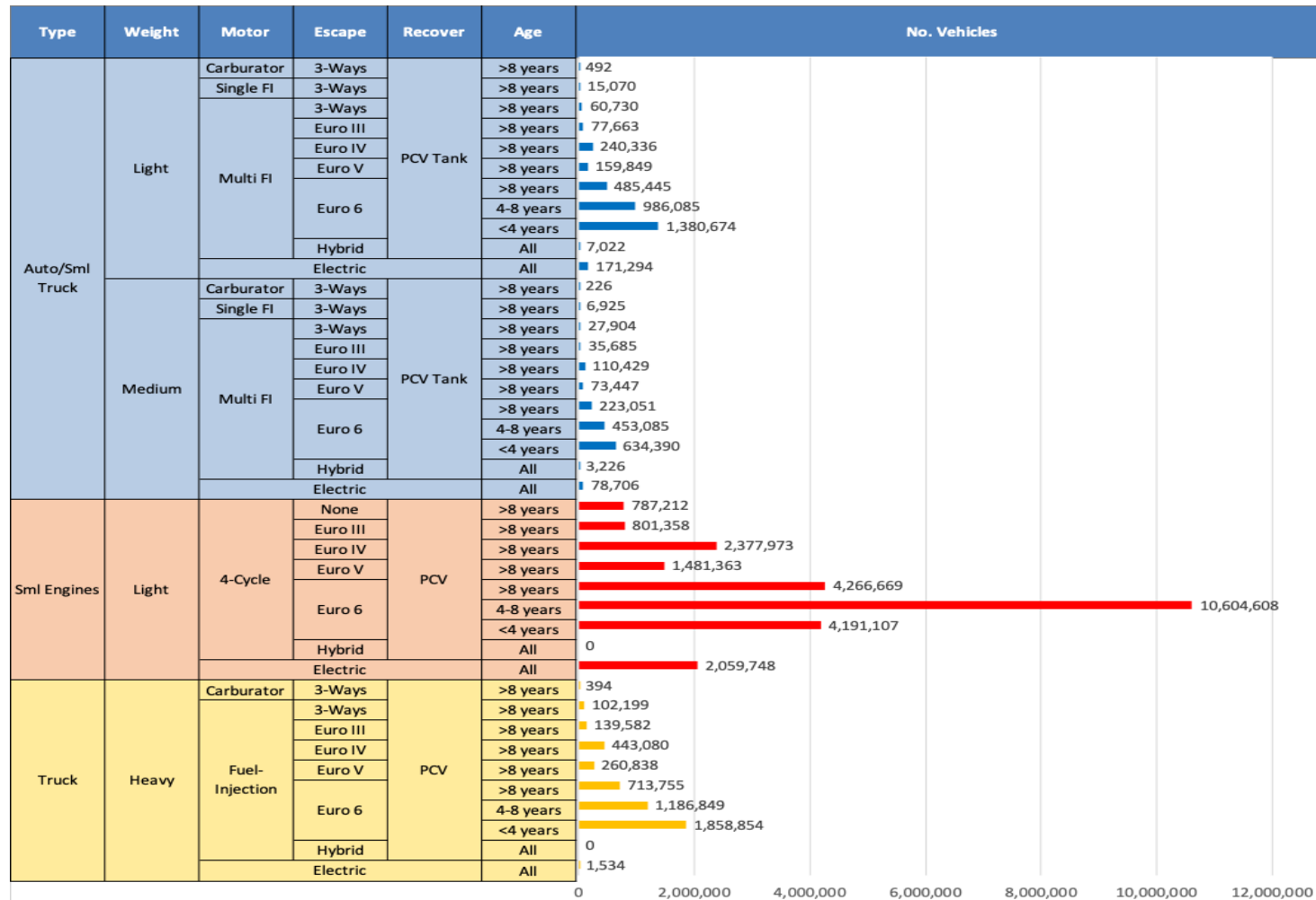
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Vietnam



Vehicle Fleet: 35,508,859
Gasoline Vehicle Fleet: 25,550,917

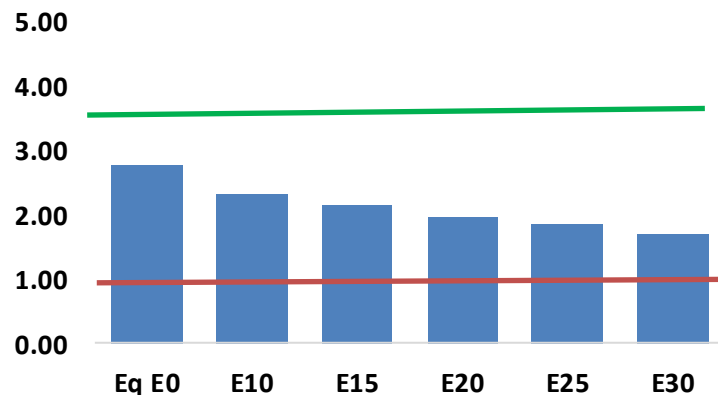
Source: MoIT

Vietnam – Vehicular Emissions

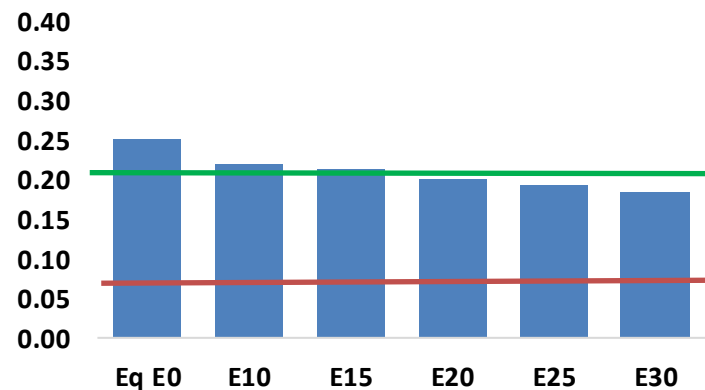
TIER III BIN 125 United States
Euro 6 Standard



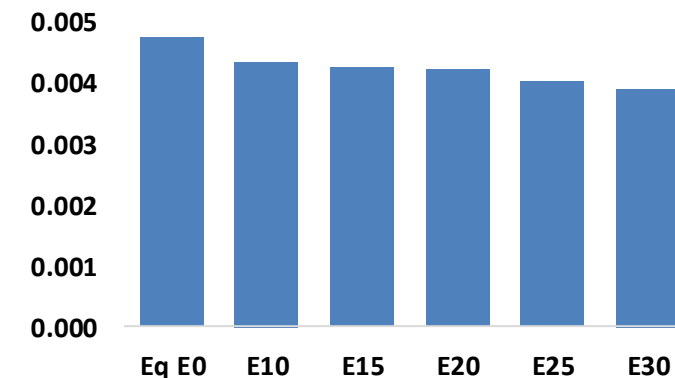
Carbon Monoxide (CO) g/km



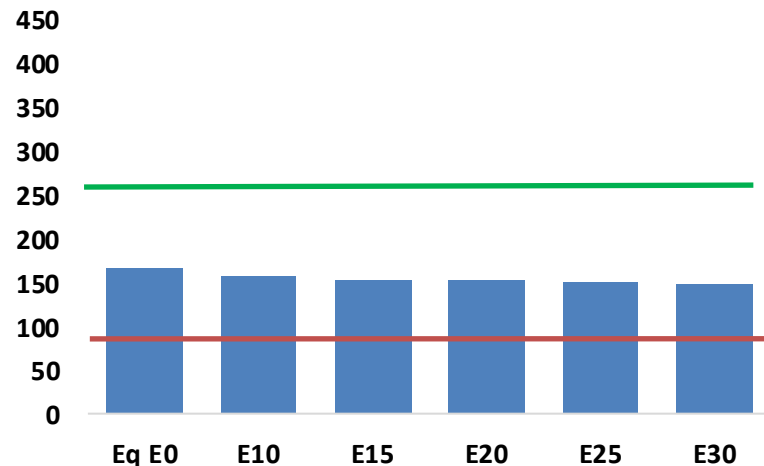
Nitrogen Oxides (NOx) g/km



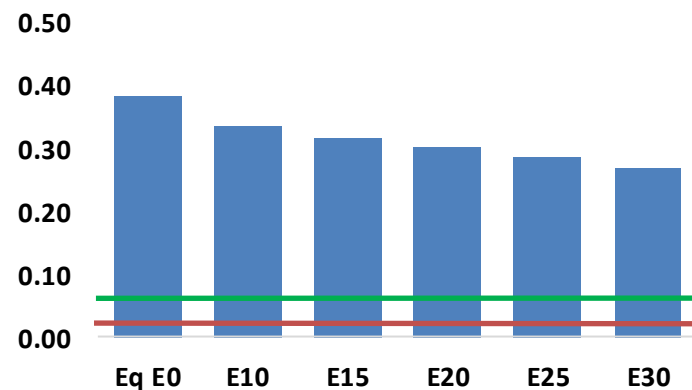
Aromatics (BTX) g/km



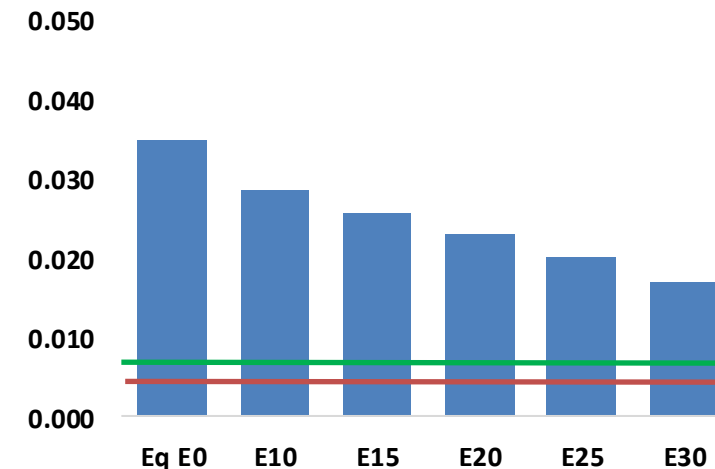
Carbon Dioxide (CO2) g/km



Volatile Organic Compounds (VOC) g/km



Particulate Matter (PM2.5) g/km



Vietnam – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,359,820	1,252,428	1,145,036	1,056,476	967,099	898,296	825,989	-16%	-29%
CO2	80,379,150	78,312,257	76,360,192	74,824,950	74,065,253	73,754,180	72,394,679	-5%	-8%
VOC	186,910	175,504	164,098	155,329	146,771	140,271	131,313	-12%	-21%
NOx	122,884	113,463	107,004	103,717	98,418	94,266	89,376	-13%	-20%
SOx	872	806	740	685	631	573	512	-15%	-28%
NH3	33,331	33,001	32,671	32,826	33,195	33,455	33,671	-2%	0%
N2O	1,350	1,342	1,337	1,343	1,371	1,386	1,402	-1%	2%
CH4	47,919	47,919	47,919	48,637	49,692	50,602	51,465	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,465	1,391	1,318	1,265	1,216	1,179	1,126	-10%	-17%
Acetaldehyde	3,770	4,327	6,348	9,668	13,171	15,051	17,784	68%	249%
Formaldehyde	14,599	14,194	16,546	19,429	20,354	22,517	24,513	13%	39%
Benzene	2,328	2,237	2,120	2,086	2,069	1,979	1,915	-9%	-11%
PM2.5	5,474	4,977	4,479	4,041	3,612	3,153	2,672	-18%	-34%
PM10	11,540	10,491	9,442	8,519	7,615	6,647	5,634	-18%	-34%

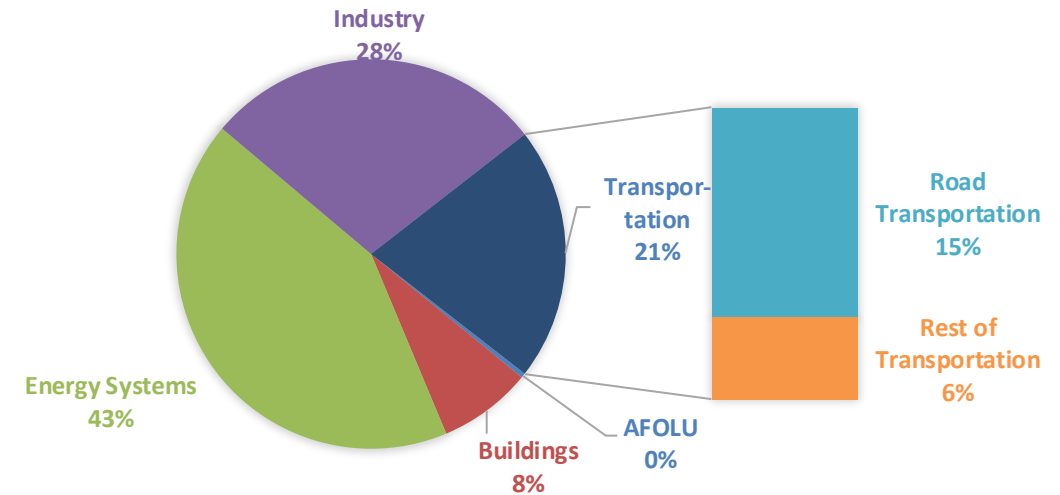
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

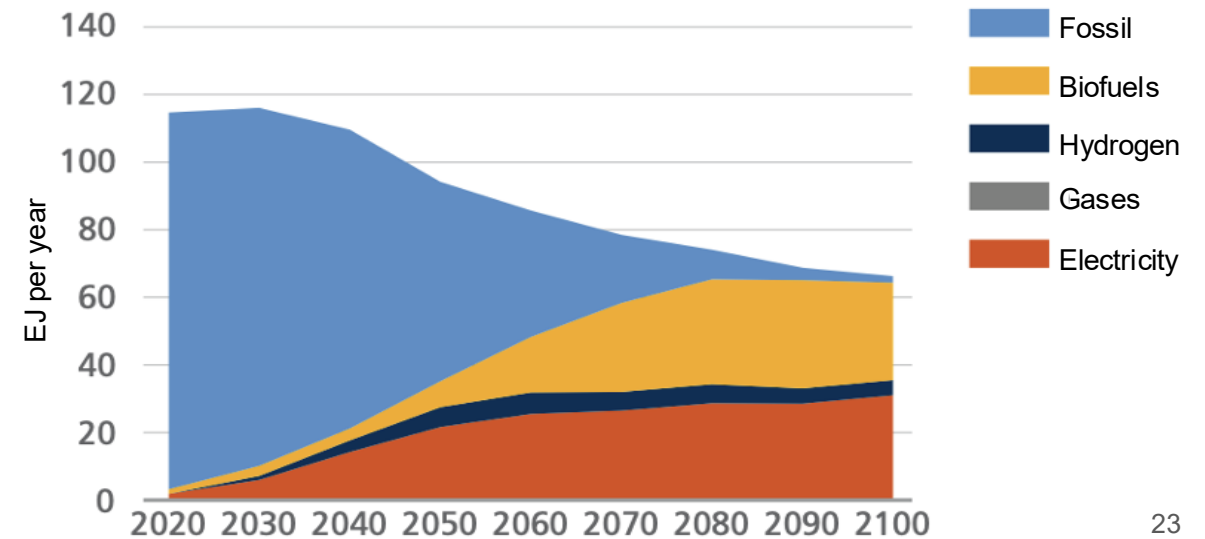
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

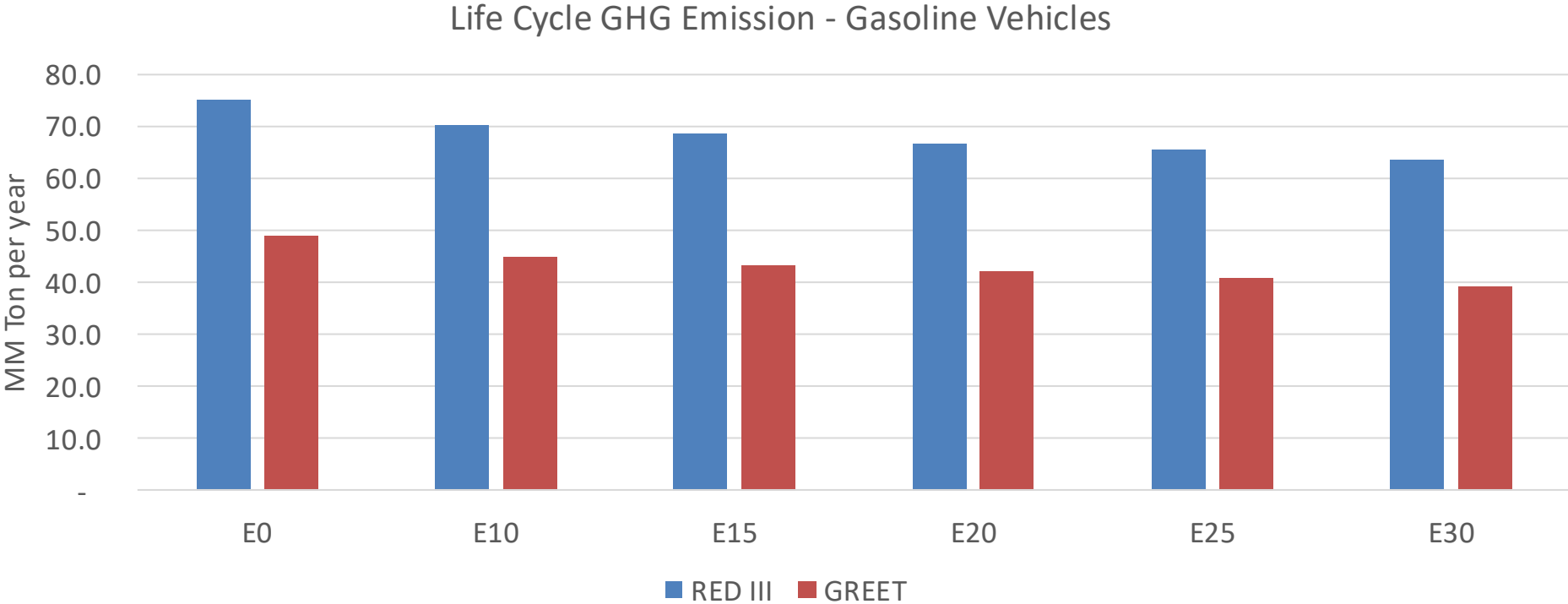
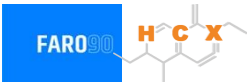
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Vietnam – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	489.9
Target @ 2035 (MMT/year)	330.6
Delta	159.3

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.6%	11.5%	14.4%	16.7%	20.0%
RED II/III	0.0%	6.5%	8.9%	11.1%	13.0%	15.5%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.6%	3.5%	4.4%	5.1%	6.1%
RED II/III	0.0%	3.1%	4.2%	5.2%	6.1%	7.3%

Source: Greet, RED II/III, climateactiontracker.org, F90